TRANSFORMING CONSTRUCTION WASTE

Recycling construction waste makes sense for this Wisconsin home builder.

BY SHANNON DELANEY AND STEPHANIE BRIEN

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Veridian Homes is the first Wisconsin home builder to adopt a construction waste recycling program.

You've managed to achieve an impeccable yard, with brilliant green grass, healthy trees, and beautiful landscaping. You recycle your cans and bottles and even participate in some backyard composting.

But your home's environmental impact started long before you moved in. In fact, your commitment to living in a greener home started when you chose your home builder. Home builders are finding a number of smart ways to lessen their environmental footprint. Specifically, construction debris recycling is proving to be a win-win choice for the builder and the environment.

In early 2007, Shawn Fahey and his wife moved into their new house in McFarland, Wisconsin, just south of Madison. The two can look out over their lawn and know that the green they see is in part due to the environmentally sound way their contractor chose to deal with their home's construction debris.

"I do a lot of outdoor activities," Fahey, a former construction manager, says. "So I'm always conscious of what I'm doing for the environment."

According to EPA, every new home built in Wisconsin creates an average of 4.38 lb of construction waste per square foot. That means a 2,000 ft^2 home would generate about 8,760 lb of building debris. Most of that material can be recycled if the home builder makes it a priority. The Faheys built their house with Veridian Homes, a family-owned home builder in south central Wisconsin that includes construction waste recycling as part of the building process in the 450 homes it builds each year.

Wood and Cardboard

Since 1999, Veridian Homes has been recycling wood and cardboard from its residential construction projects. It was the first Wisconsin home builder to adopt a construction waste recycling program. In fact, Veridian's recycling efforts have saved the equivalent of 29,000 trees, according to WasteCap Wisconsin, a nonprofit organization that assists Veridian Homes with its recycling efforts.

"Cardboard is banned from our landfills," Ralph McCall, senior project manager for WasteCap Wisconsin, says.

Construction debris recycling is proving to be a win-win situation for the builder and the environment.

"The amount of cardboard generated on a construction site really adds up, and builders can make a significantly positive impact by recycling it."

Before the Faheys moved into their house, and even before the soil was set, Veridian's environmentally conscious approach helped them to fulfill state regulations protecting the construction site against soil erosion. Many construction sites use silt fence, a barrier that prevents newly exposed soil from collecting in nearby streets or rivers. However, silt fence is not the most sustainable option.

In the early stages of Veridian Home's recycling program, wood scrap was ground and spread on the site for erosion control. It worked better than silt fence—but Wisconsin building codes require that erosion control be erected on the construction site as soon as excavation begins. Wood scrap is not generated until framing is complete—usually four to six weeks after excavation. An elaborate schedule of grinding wood from one location and transporting it to another for erosion control soon became impracticable. In addition, the wood scrap needed to be constantly replenished as workers and equipment trampled the soil, causing the ground wood to lose its effectiveness.

Wood scrap from other Veridian building projects was hauled to the Bruce Company, a local landscaper, to be partially composted and blown into long fabric tubes known as silt socks. These socks were placed along the perimeter of the Fahey building site for erosion control. They are preferable to silt fence because they trap sediment while allowing water to pass through.

"Silt fences often fail even in moderate rainfall, because the weight of the water will cause the fence to collapse," McCall says. Silt fence can also prevent easy access to the building site, and they need to be taken on and off the site throughout the day, according to Gretchen Wheat, a waste management engineer from the Wisconsin Department of Natural Resources. When construction is finished, the silt socks are cut open and the composted wood material is spread over the lot to form a base for topsoil and lawn seed. Silt fences, on the other hand, are rolled up and tossed in the trash bins.

"There is much less waste with the silt socks," McCall says. But even though silt socks proved to be more efficient than wood chips for erosion control, both options were better environmental choices than silt fence.

The wood scrap used for erosion control consists partly of engineered lumber. This is true whether the wood scrap is used in the form of compost for silt socks or in the form of wood chips. The concept of using engineered lumber for this purpose is new in Wisconsin. Originally environmentally conscious builders thought that the glue and chemicals in the engineered lumber would pollute the groundwater and the soil. So Veridian Homes, WasteCap Wisconsin, and other state partners went through state testing labs and a county case study to determine whether this was the case. The partners were able to show the Wisconsin Department of Natural Resources not only that it was impractical to separate engineered lumber from the wood scrap mix at a construction site, but also that engineered lumber could be safely used in wood chips, and in silt socks for erosion control. Veridian Homes then received the state's first-ever permit to grind engineered lumber along with dimensional lumber for use in erosion control on residential construction sites.

"About a third of the waste going into Wisconsin's landfills is construction and demolition debris," says McCall. "This permit paves the way for builders like Veridian to significantly decrease the amount of waste they are generating while also significantly benefiting the environment."

To make recycling as easy and accessible as possible, Veridian places a cardboard dumpster and a scrap wood dumpster on every new home site. Building crews separate the materials, and Pellitteri Waste Systems, a local hauling company committed to recycling, hauls the cardboard to recycling centers.

Recycling Vinyl and Drywall

"After seeing the environmental impact of recycling wood and cardboard, we began expanding our efforts to see what else we could recycle, one product at a time," says Gary Zajicek, Veridian's vice president of construction. Vinyl recycling is one of his company's newest initiatives. In 2005, Veridian partnered with the Vinyl Institute, the Wisconsin Department of Natural Resources, WasteCap Wisconsin, and Waste Management Recycle America to test different methods of recycling vinyl throughout metropolitan Madison and Milwaukee.

A 2,000 ft^2 home produces only about 150 lb of excess siding. Pete Lindabery, a plastic recycling consultant for the Vinyl Institute, says that one of the biggest problems with recycling vinyl scrap is finding a site


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Silt socks trap sediment while allowing water to pass through.

that generates enough material to make the testing of recycling methods worthwhile.

WasteCap Wisconsin and its partners continue to work with various contractors to collect vinyl scrap from construction site projects, both commercial and residential, to test recycling methods and find processors and end markets for the recycled material.

Veridian's drywall contractor works with the Drywall Scrap Company —one of the few companies in Madison that have a low-hazard exemption from the Department of Natural Resources to recycle drywall. The company uses a method called indirect land application via animal bedding. After collecting scrap drywall from Veridian's construction sites, the company transports it to county farmers for use in animal bedding. Clean drywall scrap is mixed with straw, corn stalks, and other crop residue and spread in cattle barns and


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Gary Zajicek (left), vice president of construction at Veridian, accepts the WasteCap Wisconsin R3 award from WasteCap Wisconsin Board of Directors President David Shoemaker.

feedlots. The material breaks down and mixes with manure as the cattle walk over it. The resulting mixture is then collected and applied to croplands as a fertilizer and soil amendment. The Drywall Scrap Company estimates that it is diverting close to 100 tons of drywall a month from landfills.

A Successful Recycling Program

A productive wood, cardboard, drywall, and vinyl recycling program on a residential construction site must start with the commitment of the builder, who will oversee the project, keep all materials separated, and educate the crews.

"A lot of our customers don't know about Veridian's environmental efficiency and recycling methods when they walk in the door," Zajicek says. "If your county has a local resource to help train builders and find local markets, encourage them to use it."

In 2006, Zajicek accepted a WasteCap Wisconsin R3 award for Veridian's commitment to recycling residential construction waste. And according to WasteCap Wisconsin, Veridian Homes is now recycling an impressive 62% of its construction debris, including wood, cardboard, drywall, and vinyl siding.

But Veridian is doing much more than just recycling its construction waste. Every Veridian home built is Wisconsin Energy Star (WES) certified. This means that for every ten homes Veridian builds, enough energy is saved to light one additional home. In an average year, Veridian's environmental measures will prevent almost 500,000 lb of CO_2 and more than 400 lb of nitrous oxide (N_2O) from being released into the air. These greenhouse savings also represent more than $30,000 in homeowner savings.

Owners have one chance to build a green home. After that, they will have many opportunities to establish and maintain green home traditions. The Faheys can recycle their moving boxes and packing peanuts, for example. They can become familiar with local recycling rules. They can purchase efficient appliances for their new home. When it comes time to remodel, they can donate fixtures to reuse stores and support other local markets that help make it easier to recycle construction waste.

The Faheys smile as they look through the new windows at their green lawn. They are proud to support a local builder that values the land it builds its homes on.

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For more information:

To learn more about Veridian Homes, go to www.veridianhomes.com.

For more information on WasteCap Wisconsin, go to www.wastecapwi.org.